



FLIGHT BOOKING SYSTEM

A Complete, Secure, and Automated REST API built with Java Spring
Boot

The Tech Stack

Modern tools powering the backend infrastructure.



Java Spring Boot

The core engine. It handles strict validation, processes data, and securely routes all user actions.



MySQL & JPA

Manages 10 structured database tables (Users, Flights, Tickets, Cards) seamlessly.



Global Exception Handler

A centralized controller advice that instantly catches and formats any data or logic errors.

The System Setup Flow

How Administrators prepare the system before users can buy tickets.



1. Add Airlines

Create authorized companies like Emirates or Qatar Airways.



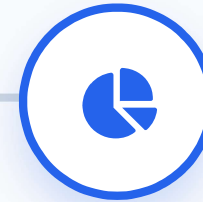
2. Add Airplanes

Assign specific planes and exact seat capacities to airlines.



3. Schedule Flights

Set departure cities, travel dates, and exact ticket prices.



4. Track Profits

Monitor total revenue generated from all completed purchases.

The Customer Journey

A fast, secure, and automated ticket purchasing process.



1. Search

Users find flights by selecting cities and desired travel dates.



2. Verify

System checks user login, flight seat availability, and card ownership.



3. Purchase

Card balance is checked and money is safely deducted.



4. Ticket Issued

Passenger data is saved and a ticket is emailed instantly.

Real-Time Notifications

Automated Email and WhatsApp integration for travelers.



Always Connected

The system uses external API endpoints to push critical updates directly to the user's phone and inbox instantly.



1. Registration

Sends a welcoming onboarding message immediately after a new account is created.



2. Tickets & Receipts

Automatically delivers the digital ticket and full payment summary upon a successful purchase.



3. Cancellations

Alerts users about flight cancellations and instantly notifies them of their processed refund.

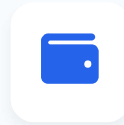
Secure Payment Processing

Handling financial transactions safely and efficiently.



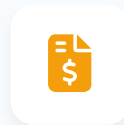
Bank-Grade Validation

The system strictly validates card ownership and verifies adequate funds before any ticket is generated.



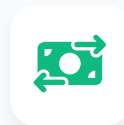
Instant Deduction

Upon approval, the exact ticket cost (plus markup) is instantly deducted from the card's balance.



Financial Logging

Every transaction is saved in the database with a clear "PURCHASE" or "CANCELLED" tag for accurate accounting.



Card Top-Ups

Users have access to a dedicated endpoint to securely add funds back into their account balance.

Dynamic Seat Inventory

Preventing overbooking in real-time.



Live Tracking

The system actively monitors the exact number of remaining seats for every scheduled flight in the database.



No Overbooking

If a flight reaches 0 capacity, the system blocks all incoming transactions and displays a "Fully Booked" error.



Auto-Restocking

If a ticket is refunded or an admin cancels a reservation, that exact seat is instantly returned to the flight's inventory.

Smart Flight Search Engine

Helping users find exactly what they are looking for.

Instead of scrolling through hundreds of random flights, our system filters and organizes results instantly based on user preferences.

Built with Java Comparators

The system uses efficient sorting algorithms on the backend to organize data before sending it to the user.



Exact Route Matching

Searches match exact Departure and Destination cities for the requested date boundaries.



Sort by Price

Organizes the search results to show the absolute cheapest flights at the very top.



Sort by Duration

Organizes the search results to show the shortest, fastest flights first.

Automated Mass Refunds

Handling real-world emergencies securely and instantly.

What happens when a flight is cancelled?

If an administrator cancels a scheduled flight, the system protects users by automatically triggering a massive database cleanup loop.



1. Locate Payments

Finds every single purchase made for that specific flight.



2. Return Funds

Refunds the exact amount back to every user's credit card.



3. Clean Up Data

Safely deletes the stranded tickets and passengers.



4. Notify

Triggers the Email API to apologize and confirm.



SYSTEM READY.

A highly scalable, secure, and professional enterprise backend.